

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

Department of Humanities & Social Sciences

End-term Examination [B. Tech. Semester I, Sections: G, H, I & J] November 2025

22HST101: Basic Economics

Duration: 2.5 hours

Max. Marks: 50

Section I: Attempt 15 questions in about 50 words each. [15 × 2 mark = 30 marks]

1. If the government decides to finance its spending on welfare schemes and economic support programs by printing additional currency rather than raising taxes, who would effectively bear this "inflation tax" in the Indian economy, and why?
2. Identify whether the following statements are related to microeconomics or macroeconomics.
a. "The Reserve Bank of India has decided to keep the repo rate unchanged to control inflationary pressures." b. "Small textile units in Gujarat report higher production costs due to increased cotton prices."
3. In a market for mangoes, the demand and supply functions are obtained as $Q_D = 100 - 2P$ and $Q_S = 20 + 3P$, respectively. Determine whether there is a shortage, surplus or equilibrium, if the price is set at ₹10 per kg.
4. Define income elasticity of demand. For an inferior good, will the income elasticity of demand be positive or negative?
5. Explain the law of diminishing marginal utility, using an example.
6. A consumer spends all her income of ₹120 on goods X and Y. The prices of X and Y are ₹15 and ₹10 per unit, respectively. Find the budget equation and plot it.
7. What is Marginal Rate of Substitution (MRS)?
8. Draw the marginal-cost and average-total-cost curves for a typical firm. How are they related?
9. You start a pizza delivery business where the monthly rent is ₹5,000 and the labour & raw material cost is ₹200 per pizza. Write the total cost (TC) function and calculate the total average cost (AC) and marginal cost (MC) when producing 50 pizzas.
10. A firm in a competitive market receives ₹50,000 in total revenue and has a marginal revenue of ₹100. What is the average revenue, and how many units were sold?
11. Give an example of government created monopoly. Is creating this monopoly necessarily bad public policy/ explain.
12. What are nominal GDP and real GDP? Explain the difference between the two.
13. What is Purchasing Power Parity (PPP)? How does PPP help in comparing living standards and price levels across countries?
14. What is fiscal policy? State any two major objectives of fiscal policy.
15. What is the Human Development Index (HDI)? State the key indicators used to measure it.
16. Define the terms-Deflation and Stagflation.

$TR = TC - MC$

Section II: Attempt 5 questions in about 100 words. [5 × 4 mark = 20 marks]

17. Draw a Production Possibility Frontier (PPF) for an economy that produces cars and computers. What happens to this frontier in case of the following events:
 - a. Advancement in technology
 - b. Improvement in education and training of the labour force
 - c. Destruction of resources due to an earthquake
18. Consider the electric vehicle (EV) market in India. For each event below, identify the affected determinant of demand or supply, state whether demand or supply increases or decreases, and illustrate the effect on equilibrium price and quantity using a properly labeled diagram.
 - a. The GOI increases subsidies on electric car purchases under the FAME policy.
 - b. Due to rising global demand for lithium, the cost of EV batteries increases.
 - c. Improvements in EV manufacturing technology reduce cost of production.
 - d. Charging infrastructure expands rapidly, improving the convenience of owning an EV.
19. The price of petrol rises from ₹100 to ₹110 per litre. As a result, the monthly consumption in a city falls from 10 lakh litres to 9.8 lakh litres.
 - a. Calculate the price elasticity of demand for petrol.
 - b. Based on your result, state whether the demand for petrol is elastic or inelastic.
 - c. Briefly explain why the demand for petrol shows this behaviour.
20. What are the conditions for consumer's equilibrium according to indifference curve analysis? Illustrate your answer with a suitable diagram.
21. Define *economies of scale* and *diseconomies of scale*. Explain why they might arise.
22. For each of the characteristics listed below, identify the market structure it best represents and provide a brief explanation to justify your answer.
 - a. Price discrimination **monopoly**
 - b. Product differentiation **monopolistic**
 - c. Collusions among firms **oligopoly**
 - d. Mergers and acquisitions aimed at increasing market dominance **oligopoly**
 - e. Free entry and exit of firms **competitive**
 - f. No close substitutes for the firm's product **monopoly**
 - g. Firms are price takers **competitive**
 - h. Significant advertising and branding **monopolistic**
23. Explain monetary policy and its tools.

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Humanities & Social Sciences
End-term Examination [B.Tech.Semester II, Sections: A, B, C, D, E, & F] May 2025

22HST101: Basic Economics

Max. Marks 50

Duration: 2.5 hours

Section A [10 x 2 marks = 20 marks]

Attempt all questions in about 40-50 words.

1. Identify whether the following statements are positive or normative. Give reason.
 - a) The government should provide free education.
 - b) An increase in taxes leads to lower consumption.
2. Explain the role of the 'invisible hand' in the functioning of a market.
3. Explain why the production possibility curve is typically concave to the origin.
4. What will happen to the demand for umbrellas if weather forecasts predict heavy rain for the next week? Explain using the concept of the demand function.
5. The price of good X increases by 10% and the demand for good Y increases by 8%. Calculate the cross-price elasticity of demand for Y, and interpret the relationship between the goods X & Y.
6. Explain the law of diminishing marginal utility and mention any two exceptions to it. *1.05*
7. 'A large-scale bakery reduces per-unit costs by purchasing ingredients in bulk.'
Briefly explain the economic concept exemplified by the above statement.
8. A firm has a budget of ₹1,000 to spend on labour and capital. The price of labour (L) is ₹100 per unit, and the price of capital (K) is ₹50 per unit. Write the isocost equation and plot it.
9. Differentiate between monopoly and monopolistic competition.
10. The government reduces income tax and increases spending on infrastructure. Identify the type of policy measure adopted and the expected economic impact.

Section B [5 x 6 marks = 30 marks]

This section contains six questions carrying six marks each. Attempt any 5 questions in about 100-150 words.

11. Describe the short-run trade-off between inflation and unemployment. Explain how policymakers might respond to high inflation vs. high unemployment.
12. Using the circular flow of income model, explain the interdependence between households and firms in a four-sector economy.
13. Explain the conditions necessary for a consumer to attain equilibrium using the indifference curve approach. Illustrate your answer with a diagram.
14. Discuss the three stages of the Law of Variable Proportions with the help of Total Product (TP), Average Product (AP), and Marginal Product (MP) curves.
15. A local bakery produces gourmet cupcakes. The bakery's total cost (TC) function is given by $TC = 1500 + 10Q$, where Q is the number of cupcakes produced. The market price (P) is Rs 30, and the total revenue (TR) function is $TR = 30Q$. Calculate: a) Average Revenue (AR), b) Marginal Revenue (MR), c) Determine the break-even level of output, d) Analyse the bakery's profit situation.
16. Explain the concept of Sustainable Development Goals (SDGs). How are they important for inclusive growth?

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MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Humanities & Social Sciences
End-term Examination [B. Tech. Semester I, Sections: G, H, I & J] November 2024

22HST101: Basic Economics

Duration: 2.5 hours

Max. Marks: 50

Section A

Section A contains 7 questions carrying 2 marks each. Attempt all questions in about 30-40 words.

1. On the *Forbes* list of the richest people in the world (as of 22 Nov 2024), Elon Musk ranks at the top with a net worth of \$ 314.4 billion. Does this 'richest man in the world' face scarcity, or does scarcity only affect those with more limited incomes and lower net worth?
2. If the price of coffee rises from ₹45 to ₹55 per pack, and as a result the consumer's demand for tea increases from 600 to 800 packs. Find out the cross-price elasticity of demand of tea for coffee.
3. What is the difference between *Marginal Utility* and *Marginal Rate of Substitution*?
4. The following table gives the total output as a function of labour units used. Does it indicate a situation of increasing returns, diminishing returns or constant returns? Explain.

LABOR	0	1	2	3	4	5
TOTAL OUTPUT	0	5	9	12	14	15
5. How are average and marginal costs related in the short run? Show with the help of a diagram.
6. What do *Economies of Scope* mean? Give an example.
7. Define: i) Stagflation and ii) Hyperinflation.

Section B

Section B contains 7 questions carrying 6 marks each. Attempt any 6 questions in about 100 words.

8. Suppose a hypothetical household has ₹500 to spend per month on goods *X* and *Y*. If the price of *X* is ₹5 and the price of *Y* is ₹10 -
 - a. Plot the household budget constraint.
 - b. Assume that the household splits its income equally between *X* and *Y*. Show where the household ends up on the budget constraint.
 - c. Suppose the household income doubles to ₹1,000. After the change, the household spends ₹200 on *Y* and ₹800 on *X*. Does this imply that good *X* is a normal or an inferior good? What does it imply for good *Y*?
9. Elucidates how the least cost factor combination of producing a given level of output is obtained using the concept of isoquant and isocost line. Use diagram.
10. Compare and contrast the market structures representing perfect competition, monopoly, monopolistic competition, and oligopoly. Discuss how each market structure affects pricing, output, and efficiency. Provide examples of industries for each of them.
11. Explain the concepts of Purchasing Power Parity (PPP), the Human Development Index (HDI), and the Happiness Index. How do these measures provide a more comprehensive understanding of a country's development compared to traditional economic indicator, GDP?
12. A brick company manufactures a standard stone block for the building industry. The production capacity for the year is 100,000 standard blocks. The selling price per block is \$1.60, variable costs are \$0.60 per brick and fixed costs are \$60,000 per annum. Calculate the break-even level of output and the total revenue if the company breaks even.
13. Explain why the long-run average cost (LAC) curve is typically U-shaped.
14. Explain the meaning and tools of fiscal and monetary policies. How do these policies influence economic stability and growth?

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Humanities & Social Sciences

End-term Examination [B. Tech. Semester II, Sections: A, B, C, D, E & F] May 2024

22HST101: Basic Economics

Duration: 2.5 hours

Max. Marks: 50

Section A

This section contains twelve questions carrying two marks each.

Attempt any TEN questions. [10 x 2 marks = 20 marks]

1. Why should policymakers think about incentives?
2. Popeye's income declines, and he buys more spinach. Is spinach an inferior or a normal good? What happens to Popeye's demand curve for spinach?
3. How does a surplus or shortage occur in a market? Illustrate with the help of demand and supply curves.
4. Seema has decided always to spend one-third of her income on clothing. What is her income elasticity in terms of clothing demand?
5. Can Indifference curves intersect each other? Give reason.
6. Anu's budget line relating good X and good Y has an intercept of 50 units of good X and 20 units of good Y. If the price of good X is Rs.12, what is Anu's Income and the price of good Y?
7. A firm's short-run production function is given by $Q=10L-0.5L^2$ where Q is the quantity of output and L is the quantity of labour. Determine the average product of labour (AP_L) when $L=8$.
8. Define the Marginal Rate of Technical Substitution (MRTS).
9. Suppose you and your roommate started a burger delivery service on campus. List some of your fixed and variable costs.
10. What are the characteristics of a competitive market? Which of the following drinks do you think is best described by these characteristics?
a) tap water b) bottled water c) cola
11. What is Purchasing Power Parity (PPP)? How does PPP help in comparing living standards and prices between different countries?
12. Why is it desirable for a country to have a large GDP? Give an example of something that would raise GDP and yet be undesirable.

Section B

This section contains seven questions carrying five marks each.

Attempt any SIX questions. [6 x 5 marks = 30 marks]

13. Imagine that you are a policymaker trying to decide whether to reduce the rate of inflation. To make an intelligent decision, what would you need to know about inflation, unemployment, and the tradeoff between them?
14. What are the conditions of the consumer's equilibrium under the indifference curve analysis? Illustrate.
15. Explain the Law of Variable Proportion. Define the stage of rational production decision in the short run.
16. Compare and contrast the market structures of monopoly, monopolistic competition, and oligopoly. Highlight the differences in terms of market power, barriers to entry, product differentiation, pricing behaviour, and efficiency.
17. Shubhangi's Juice shop has the following cost schedules:

Quantit y	Total Variable cost	Total Cost
0	0	30
1	10	40
2	25	55
3	45	75
4	70	100
5	100	130
6	135	165

- a) Calculate total fixed cost (TFC), average variable cost (AVC), average fixed cost (AFC), average total cost (ATC), and marginal cost (MC) for each quantity.
- b) How are the marginal cost and average total cost curves for a typical firm related?
18. A small manufacturing company produces perfume bottles. The company's total cost (TC) function is given by $TC=1000+20Q$, where Q is the number of perfume bottles produced. The market price (P) is Rs 50, and the total revenue (TR) function is $TR=50Q$. Calculate: a) Average Revenue (AR); b) Marginal Revenue; c) determine the break-even level of output d) analyse the firm's profit situation.
19. Define economies of scale and diseconomies of scale. Explain why they might arise.

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Humanities & Social Sciences
End-Term Exam [B. Tech. I Semester: Sections A B C D & E], December 2019
Basic Economics (HIST 102)

Duration: 2.5 hours

Max. Marks: 50

Note: Draw graphs wherever necessary

Section A: All five questions are compulsory.

[5 x 5 marks = 25 marks]

1. Define Economics. Discuss the basic questions that economic principles answer.
2. Describe and draw consumer's equilibrium as explained by indifference curves theory of consumer behaviour.
3. Explain the law of variable proportions and the three stages of production. Which is the stage of economic relevance and why?
4. Distinguish between various types of market structures on the basis of their characteristics and give examples from practical world.
5. 'Efficient investment decisions determine the long run survival of the firm in the industry'. In view of the above statement discuss various criteria used for the selection of a project.

Section B: Attempt any five questions.

[5 x 5 marks = 25 marks]

6. Write brief notes on any two of the following:
 - a. Demand Pull and Cost Push Inflation.
 - b. Objectives of Monetary policy.
 - c. Instruments of Fiscal policy.
 - d. Balance of Payments.

(2.5+2.5)
7. State the alternative objectives of a firm. What is meant by breakeven point? Draw graph showing breakeven level of output.
Given the following demand and total cost equations: $P = 28 - 0.0008Q$; $TC = 120,000 + 0.0006Q^2$ where P is price, Q is output and TC is total cost, calculate profit maximising Q. (1+2+2)
8. Explain why the following statements are incorrect:
 - (a) Average costs are minimised when marginal costs are minimum.
 - (b) As fixed costs never change, AFC is constant at each level of output.
 - (c) Projects are selected when Benefit-Cost Ratio is less than 1.
 - (d) Demand is perfectly inelastic under monopoly.
 - (e) Entry of new firms is free and easy in oligopoly market structure.

(1+1+1+1+1)
9. Describe demand function and supply function. Draw the effect of the following:
 - (a) Technological advancement on the supply of laptops (in terms of supply curve). ☒
 - (b) Increase in GDP on the demand for branded shoes (in terms of demand curve). ☒
 - (c) Decrease in the price of coffee on the demand for and supply of coffee (in terms of both demand and supply curves) ☒

(2+3)
10. Neatly draw and label the long run average and marginal cost curves. What is meant by optimum output, optimum plant and optimum firm. (4+1)
11. (a) A manufacturing firm uses two inputs labour (L) and capital (K). The firm produces and sells a given output. You have the following information $P_L = \$20$, $P_K = \$15$, $MP_L = 10$ and $MP_K = 15$. Is the firm operating efficiently? Why or why not?
(b) A firm is currently employing 20 workers, the only variable input, at a wage rate of \$60. The average product of labour is 30 and TFC is \$3600. What is AVC, AFC and AC? (2.5+2.5)

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
 Department of Humanities & Social Sciences
End-Term Exam, April 2015: B. Tech. II Semester Sections F G H I & J
Subject: Basic Economics (HST 102)

Time: 2 hrs.

Note: Draw graphs wherever necessary.

Marks: 40

I Attempt any five questions

1. Differentiate between **any two**
 - a. Cash Reserve Ratio and Statutory Liquidity Ratio
 - b. Inflation and Stagflation
 - c. GNP and GDP
 - d. Balance of Payments and Balance of Trade

(2.5+2.5)
2. What is meant by capital budgeting? Briefly explain the merits and demerits of various criteria used for project selection. (2+3)
3. 'A perfectly competitive firm is a price taker and a monopoly firm is a price maker'. In view of this discuss the features of various type of market structures citing practical examples. (4+1)
4. Draw and describe circular flows of income in a two sector and four sector economy. (2.5+2.5)
5. Examine the properties of an isoquant. Given the iso cost line explain producer's equilibrium. (2.5+2.5)
6. Answer **any two**
 - a. Due to changes in demand or supply, equilibrium price and quantity has increased in the market. What change in market forces might have caused this, show using demand and supply graph.
 - b. Due to changes in demand or supply, equilibrium price has increased and quantity has decreased in the market. What change in market forces might have caused this, show using demand and supply graph.
 - c. Due to changes in demand or supply, equilibrium price and quantity has decreased in the market. What change in market forces might have caused this, show using demand and supply graph.

(2.5+2.5)

II Attempt all five questions

1. A firm is currently employing 20 workers, the only variable input, at a wage rate of ₹ 60. The average product of labour is 30, the last worker added 12 units to total product, and TFC is ₹ 3600. What is (a) Marginal Product; (b) Average Variable Cost; (c) Average Total Cost.
2. The demand function for good X is $Q_X = 2,50,000 - 500P_X - 1.5M - 240P_R$ where P_X is price, M is income & P_R is price of a related good Y. Given the values of $P_X = ₹ 200$; $M = ₹ 60,000$; $P_R = ₹ 100$
 - (a) Calculate point price, point income and point cross elasticity of demand for good X
 - (b) Is good X normal, superior or inferior and why?
 - (c) Are goods X and Y substitutes or complement?
3. Given the following total revenue and total cost functions
 $TR = 50Q$
 $TC = 10000 + 30Q$
 - (a) Determine the breakeven rate of output.
 - (b) Determine the output rate necessary to earn a profit of Rs. 20000.
4. The equilibrium price in a perfectly competitive market is Rs.10. The marginal cost function is given by $MC = 4 + 0.2Q$ & the firm is presently producing 40Q. To maximize profits, should Q be increased or decreased? Explain your answer.

Given $TC = 2000 + 15Q - 6Q^2 + Q^3$; At $Q=2000$ and at $Q = 10000$, what is TFC and AFC?

(3+3+3+3+3)

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
 Department of Humanities & Social Sciences
 End-Term Exam, May 2014: B.Tech. II Semester Sections A, B, C, D & E
 Subject: *Basic Economics (HST 102)*

Time 2 hrs.

Marks: 40

Note: Draw graphs wherever necessary.

I Attempt any **five** questions

1. Differentiate between monetary and fiscal policies. List the objectives and instruments of these policies. (1+2+2)
2. What is meant by capital budgeting? Explain the merits and demerits of various criteria used for project selection. (2+3)
3. 'Competition increases consumer welfare by reducing prices'. In the light of this statement compare and contrast different market structures. Why are prices lowest under perfect competition and highest under pure monopoly? (3+2)
4. Briefly describe:
 (a) Break-even level of output
 (b) Properties of an isoquant (2.5+2.5)
5. 'Cost curves are inverse of product curves', how and why? (5)
6. Write short notes on any **two**
 (i) Price elasticity of demand
 (ii) Balance of payments
 (iii) Causes of inflation
 (iv) Economies of scale (2.5+2.5)

II Attempt all **five**

1. If total cost function of a competitive firm is $TC = Q^2 - 10Q + 28$ and market price is 20 calculate short run profit maximizing output. Also calculate the total profits earned.
2. The demand function for good X is $Q_X = 2,50,000 - 500P_X - 1.5M - 240P_R$ where P_X is price, M is income & P_R is price of a related good Y. Given the values of $P_X = 200$; $M = 60,000$; $P_R = 100$
 (a) Calculate price, income and cross elasticity of demand for good X
 (b) Is good X normal, superior or inferior and why?
 (c) Are goods X and Y substitutes or complements?
3. Given the total cost function $TC = 1000 + 200Q - 9Q^2 + 0.25Q^3$
 Determine the equations for TVC, TFC, AFC, AVC, AC & MC
4. Distinguish between normal profit and abnormal profit with an example. Is normal profit equal to the opportunity cost of the entrepreneur?
5. 'The shutdown point for the firm in the short run occurs when the price equals AVC.' Explain? Draw graph showing shutdown level of output.

(3+3+3+3+3)



MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
 Department of Humanities & Social Sciences
 End-Term Exam, November 2013: B. Tech. I Semester Sections E, G, H I & J
 Subject: Basic Economics (HST 102)

Time: 2 hrs.

Marks: 40

Note: Draw graphs wherever necessary.

I Attempt **any five** questions:

1. Define capital budgeting. Throw light on commonly used criteria of project selection. If market rate of interest is 10% and internal rate of return (IRR) of Project A is calculated as 12% will the project be selected? Explain your answer. (1+2+2)
2. "Too much money chases too few goods." Explain the concept of inflation in light of this statement. What are its causes and effects? (2+3)
3. What are the alternative objectives of a firm? Do you agree that profits can be maximised only at the cost of sales, explain. (3+2)
4. Giving examples from practical world compare different market structures. Why is perfect competition an ideal market condition? (4+1)
5. What is meant by opportunity cost? Explain the relationship between average and marginal cost with the help of short run cost curves. (2+3)
6. Briefly define **any two**:
 - a. Fiscal policy and monetary policy.
 - b. Economic growth and economic development.
 - c. GNP and GDP.
 - d. Liberalisation, privatisation and globalisation.

(2.5+2.5)

II Attempt **all five**:

1. A consumer consumes two goods, cold drink and pizza. Total money income spent is Rs. 550. The price per unit is Rs. 50 for cold drink and Rs. 80 for pizza. Find the utility maximizing units of cold drink and pizza.

Units of cold drink	Total utility of cold drink	Units of pizza	Total utility of pizza
1	35	1	72
2	65	2	136
3	90	3	192
4	110	4	240
5	125	5	280
6	135	6	312

2. A firm's production function is $Q = 3K^{0.5}L^{0.5}$; where capital (K) is fixed at 25 units in the short run. The price of capital is Rs. 30 per unit and the price of labour (L) is Rs. 5 per unit. Determine TFC, TVC and TC when Q is 45 units.
3. If fixed cost is Rs. 24000; variable cost is Rs. 8 per unit and selling price is Rs. 40 per unit, find breakeven level of output and total revenue. If profit target is Rs. 40000, find the output required.
4. You have following information about ABC Associates:
 $MP_L = 10$, $MP_K = 15$, $P_L = Rs. 20$, $P_K = Rs. 15$; is ABC being run efficiently? If not why and how can you improve its efficiency?

Prove that the marginal revenue equals the price of the product for a perfectly competitive firm with a numerical example.

(3+3+3+3+3)



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MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

Department of Humanities & Social Sciences

End-Term Exam, 2012: B. Tech. I Semester Sections A B C D & E

Subject: Basic Economics

Time 2 hrs.

Marks: 40

Note: Draw graphs wherever necessary.

I Attempt any five questions.

1. Define supply. Discuss the law of supply. What is the impact of an increase in supply of a good/service on its price? (1+2+2)
2. What is meant by Marginal Utility? Explain the law of equi-marginal utility. (2+3)
3. Distinguish between the concepts of GDP and GNP. How is national income measured? (2+3)
4. 'A firm's hold on its product price is inversely related to the level of competition in the market'. In the light of this statement distinguish between various types of market structures with examples. (5)
5. Explain the meaning of capital budgeting. Discuss various criteria used for the selection of a project. (2+3)
6. Examine the alternative objectives of a firm. 'Profits can be maximised at the cost of sales and vice-versa', explain. (3+2)

II Attempt all five

1. Using graphs show what would be the outcome on the price and quantity of the following, if *ceteris paribus*:
 - a. Investment in gold increases (gold prices)
 - b. Reduction in imports of crude oil (petrol prices)
 - c. Electricity market becomes competitive (electricity prices)
2. Three engineering graduates decide to open an enterprise, and all three devote their full time to its management. What cost would you assign to their time? Is this an explicit or implicit cost?
3. For each of the following equations, determine whether demand is elastic, inelastic or unitary elastic at the given price.
(a) $Q = 100 - 4P$ at $P = \text{Rs. } 20$ (b) $Q = 1500 - 20P$ at $P = \text{Rs. } 5$ (c) $P = 50 - 0.1Q$ at $P = \text{Rs. } 20$
4. Given the total cost function $TC = 1000 + 200Q - 9Q^2 + 0.25Q^3$
Determine the equations for TVC, TFC, AFC, AVC, AC & MC
5. Given the following TR & TC functions
 $TR = 50Q$ $TC = 10,000 + 30Q$
 - (a) Determine the break-even level of Q
 - (b) Determine the level of Q necessary to earn a profit of Rs. 20,000

(3+3+3+3+3)

